

Project Lifeline

A Content-Addressed, Capability-Governed Delay-Tolerant Mesh
for Offline, End-to-End-Encrypted Emergency Communication

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Abstract

Disasters remove exactly the infrastructure that messaging depends on: towers, backhaul, power, and the reachable servers that hold our identities and authorize our traffic. *Project Lifeline* is a decentralized, offline-first, end-to-end-encrypted messenger built as a Delay/Disruption-Tolerant Network (DTN) [9] that keeps working while partitioned. It composes a self-sovereign identity layer, a content-addressed bundle with a Merkle manifest, a store-carry-forward router using binary spray-and-wait with custody transfer, forward-secret sealed-sender encryption, a Kademlia overlay for online discovery, and pluggable bridges to existing ecosystems (Nostr, Meshtastic). This paper records the system’s architecture and its four principal design contributions: (i) a *residual differential transfer* framework that unifies several bandwidth optimizations as $\Delta = Q - \mathbb{E}[Q | R]$; (ii) *manifest-scoped HAVE swarm fetch*, a BitTorrent-over-DTN gap-fill that completes an object from many providers over many carriers with no duplicate traffic; (iii) *source-attributed custody reputation* for passively detecting black-hole relays without weakening the delivery guarantee; and (iv) *attenuable egress capabilities* with *service-class* tiers, a macaroon/Biscuit-style authorization token that governs internet access through a gateway and—crucially for a mesh—is verified fully offline. We give a security analysis mapped to a standard threat taxonomy, describe our evaluation methodology and the benchmarks the DTN literature expects, and enumerate the open problems (independent cryptographic review, comparative evaluation on standard mobility traces, and quota/revocation for egress) that gate a production deployment.

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1 Introduction

The design center of most messaging systems is a reachable server. In a disaster—an earthquake, a flood, a grid failure, a deliberate shutdown—that assumption is precisely what fails. Project Lifeline targets the regime where connectivity is *intermittent, partitioned, and heterogeneous*: some nodes share a short-range radio for minutes, some carry data physically between partitions, and a lucky few briefly reach the wider internet.

We take the position, following the end-to-end argument [36] and the DTN architecture [9, 38, 8], that reliability, security, and authorization must live at the *endpoints and in the data they carry*, never in the links or in a server one must reach. Every design decision in Lifeline is a consequence of that stance.

Contributions. This paper documents:

1. A layered architecture (§3) whose seams are narrow traits, so a new radio, overlay, or external network is a single implementation.
2. Four named contributions: the RESIDUAL DIFFERENTIAL TRANSFER framework (§11), MANIFEST-SCOPED HAVE SWARM FETCH (§6), SOURCE-ATTRIBUTED CUSTODY REPUTATION (§5.3), and ATTENUABLE EGRESS CAPABILITIES with SERVICE-CLASS tiers (§10).
3. A security analysis mapped to a standard threat taxonomy (§12) and an evaluation methodology grounded in the DTN benchmarking literature (§13).

Status and honesty. Lifeline is an implemented research system (a Rust workspace of 15 crates, ~200 automated tests, and an acceptance simulation held to a $\geq 95\%$ delivery target). It has *not* undergone independent cryptographic review; where a guarantee is aspirational or a mechanism is not yet wired into the live node, we say so explicitly.

Layer	Concern	Key abstraction / crate
L1 Identity	self-sovereign naming	ADDRESS= BLAKE3(pk)[..16] (<code>proto</code> , <code>core</code>)
L2 Bundle	sealed unit of transfer	BUNDLE, MANIFEST, content id (<code>proto</code> , <code>core</code>)
L3 Discovery	find peers / content	Kademlia DHT, geocast, time-sync (<code>dht</code> , <code>geo</code> , <code>timesync</code>)
L4 Routing	store-carry-forward	spray-and-wait, custody, attribution (<code>router</code> , <code>engine</code>)
L5 Application	messages, groups, gateway	sealed-sender, groups, egress (<code>core</code> , <code>inet</code>)

Table 1: The five layers and where each lives in the Rust workspace. The `INTERFACE`, `EXTERNAL-NET`, and `ACCESSPOLICY` traits are the extension seams: a new carrier, bridged network, or egress policy is one `impl`.

2 System and Threat Model

Network model. The network is a time-varying graph whose edges (a BLE contact, a Wi-Fi Aware link, a LoRa hop, an ultrasonic burst, an internet tunnel, a bridged relay) appear and vanish. There is no assumption of a contemporaneous end-to-end path; delivery is by *store-carry-forward*. Each edge type is an `INTERFACE` with a maximum transmission unit and a throughput class.

Adversary. We assume a Dolev–Yao network adversary that can observe, drop, replay, reorder, and inject on any link, plus a fraction of malicious *participants* who follow the wire format but misbehave (drop bundles they accept, forge control messages, or attempt to spam). We assume the endpoints’ devices and long-term secret keys are not compromised; the standard caveats of seizure and coercion apply and are discussed in §12. We explicitly do *not* claim resistance to global traffic analysis [43, 33].

Goals. In priority order: (1) *life-safety delivery*—an SOS must move even under spam pressure; (2) *confidentiality and integrity* of message content and, as far as possible, of metadata; (3) *availability* under partition; (4) *augmentation*, not replacement, of existing networks.

3 Architecture

Lifeline is organized as five layers, each a thin trait boundary (Table 1). The invariant that makes the system tractable is that *relays are dumb*: an intermediate node forwards an opaque, sealed BUNDLE and can neither read it nor is trusted to have processed it correctly. All correctness is re-established at the recipient.

In Lifeline. The narrow waist is the sealed BUNDLE, mirroring the IP hourglass. `transport::Interface` abstracts a carrier; the engine drives every attached interface concurrently and deduplicates by bundle id, so the same message riding BLE *and* LoRa is stored once. Adding Meshtastic or Nostr required only an `ExternalNet` implementation, not a change to the core.

4 Identity and the Bundle

4.1 Self-sovereign identity

A node generates an Ed25519 signing key and an X25519 key-agreement key on first launch. Its network ADDRESS is `BLAKE3(sign_pub)[..16]` —a 128-bit fingerprint of the public key. There is no registrar, phone number, or email: the key *is* the identity, which is what lets identity survive when servers do not. Addresses double as the routing destination and the Kademia key, so the same 128-bit space underlies naming and discovery.

4.2 Content addressing and the manifest

Large objects are split into fixed-size blocks, each named by its BLAKE3 hash —its *content id* (CID) [3]. An ordered MANIFEST (itself named by a root CID over the tuple of length, block size, and block CIDs) is a Merkle description of the whole object [24]. This buys three properties for free:

- **Intrinsic integrity** — a received block is trusted only if it hashes to the requested CID; substitution is detected, not assumed away.
- **Global deduplication** — identical content yields identical CIDs, so a block is carried once regardless of how many messages reference it or how many carriers deliver it.
- **A definition of “done”** — the object is complete iff the set of manifest CIDs the store is missing is empty; reassembly then verifies the whole object end-to-end.

For cross-partition resilience a bundle may additionally be protected with Reed–Solomon erasure coding [34], so any k of n shards reconstruct it even if some never arrive.

5 Routing

5.1 Store-carry-forward with binary spray-and-wait

Lifeline routes with binary spray-and-wait [41], a controlled-replication scheme that bounds the number of copies of a bundle to a budget L : a node holding $c > 1$ copies hands half to a new contact and keeps half; a node holding one copy waits to deliver directly. This gives most of the delivery probability of epidemic routing [42] at a small fraction of the transmission overhead—the metric that matters on battery- and spectrum-limited devices.

5.2 Priority and postage

Bundles carry a `PRIORITY` $\in \{\text{SOS, Alert, Normal, Bulk}\}$ that orders scheduling. To resist flooding without ever delaying life-safety traffic, Normal and Bulk bundles must carry Hashcash-style proof-of-work “postage” [2] whose difficulty scales inversely with priority; SOS is exempt (difficulty 0). A relay verifies postage before admitting a low-priority bundle, so a spammer pays $O(2^k)$ hashes per message while a genuine SOS pays nothing.

5.3 Source-attributed custody reputation

Custody transfer [38] lets a committed custodian accept responsibility for a bundle so an upstream node may free its copy. This creates a *black-hole* risk: a malicious custodian accepts custody and

silently drops the bundle. Detecting misbehavior in a DTN is hard because there is no path and no timely acknowledgement.

Our contribution here, `SOURCE-ATTRIBUTED CUSTODY REPUTATION`, exploits a structural fact: delivery receipts are sealed to the *original sender*, so only the source can verify delivery. The source keeps a ledger mapping bundle $\rightarrow$ custodian at each custody hand-off; when a sealed delivery receipt arrives, it credits the custodians on that bundle’s path; a custodian that repeatedly takes custody of bundles that never deliver (beyond a grace count) is demoted. Demotion only ever *re-routes around* a suspected node when an alternative exists—it never drops traffic—so the $\geq 95\%$ delivery target is preserved even if the heuristic is wrong.

In Lifeline. Implemented in `router::attribution::ForwardLedger`. It is passive (never alters store/forward) and source-local (no global reputation gossip to poison), which sidesteps the Sybil/eclipse amplification that plagues shared reputation systems (§12).

6 Reliable Multi-Carrier Transfer and Swarm Fetch

The disaster requirement is that an object ride *every* available carrier at once and the recipient end up with *all* of it, *exactly once*, over links that are lossy and intermittent. The internet’s designers solved the pieces of this long ago, and content addressing (§4.2) already supplies no-duplication and integrity while the manifest supplies a completeness predicate. The remaining gap is *multi-source* acquisition.

MANIFEST-SCOPED HAVE SWARM FETCH closes it, adapting the BitTorrent swarm [10] to a DTN. For an object described by a manifest, the fetcher:

1. sends each candidate provider a `HaveQuery` listing its still-missing CIDs; a provider answers `HaveReply` with the subset it holds (a HAVE bitmap scoped to one object, not the whole store);
2. requests each missing block from a provider known to hold it, with the choice offset by a per-fetch rotation counter and the block index, so distinct blocks are pulled from distinct providers in the same round;
3. on retry, rotates each still-missing block to a *different* provider, routing around a provider that has gone dark or is a black hole.

Because at most one provider is asked per block per round, there is no duplicate traffic; because blocks are content-addressed, any duplicate that does arrive is idempotent; and because the missing set is monotone non-increasing, completion is provable. This is the DTN analogue of a rarest-need, multi-source, carrier-agnostic download.

In Lifeline. `NodeEngine::fetch_content_swarm`. Verified by two tests: two providers each holding a disjoint half of the blocks (completion requires combining both) and a primary provider that goes dark after first contact (completion via the surviving provider). For *whole-store* synchronization between peers we instead use range-based set reconciliation [27, 17] (crate `reconcile`), chosen over IBLTs for its deterministic, no-silent-omission behavior.

7 Cryptographic Design

Lifeline composes audited primitives—Ed25519, X25519, HKDF-SHA256, XChaCha20-Poly1305, BLAKE3 [30], and Argon2id [5] for passphrase-wrapped key backup—and invents no new primitive.

Sealed-sender forward-secret envelope. A message is an ephemeral-static sealed box (X25519 + HKDF + XChaCha20-Poly1305) to the recipient, with the *sender’s signature placed inside the sealed plaintext*, so relays learn neither content nor sender. Forward secrecy in the present design comes not from a live ratchet—impractical when a sender may be offline for days—but from a ring of *rotating prekeys* that bounds the window a seized long-term key can decrypt. This is a deliberate, documented weakening relative to a full Double Ratchet [32]; the X3DH-style asynchronous agreement [22] is the design Lifeline approximates for the store-and-forward setting.

Groups and onion routing. Groups use sender-keys (each member distributes a symmetric sending key sealed to the others), so a message is encrypted once and authenticated per sender. An optional onion layer [13] wraps a bundle so each hop learns only the next, hiding the communication graph from any single relay.

8 Discovery: Overlay, Geocast, and Time

Kademlia overlay. When some nodes have connectivity, Lifeline builds a Kademlia DHT [23] over the 128-bit address space (k -buckets, iterative α -parallel FIND_NODE/FIND_VALUE) for locating mailboxes and content providers. The DHT is transport-agnostic (a DHTRPC request/response seam) and is explicitly *not* relied upon for offline operation.

Geocast. Geohash [28] maps a region to a base-32 cell; a *region-derived key* lets a sender address “everyone in cell g ” by deriving a deterministic keypair from the geohash. We are candid that geocast confidentiality is weak by construction (the region commitment is reversible), and document it as an addressing convenience, not a privacy mechanism.

Differential time synchronization. With no reachable NTP server [26], nodes agree on time by gossiping stratum-tagged beacons and applying a bounded-slew median correction. TTL gating on relays requires the caller to authenticate beacons first; this is enforced by API contract and noted as a hardening dependency.

9 Bridging Existing Ecosystems

Lifeline augments rather than replaces. The EXTERNALNET trait plus a BRIDGEINTERFACE adapter let any external network appear to the engine as just another carrier. Two bridges are implemented: a live WebSocket client to Nostr relays [29] (using a domain-separated bearer subkey so the Nostr identity is unlinkable to the mesh identity), and a Meshtastic adapter speaking the ServiceEnvelope/MeshPacket protobuf over MQTT [25]. Both features are off by default so the base build stays lean.

10 Capability-Governed Internet Egress

A node with real internet can act as an *exit* for others: the mesh becomes the access network, that node the gateway, over a store-and-forward request/ response proxy (not a live socket, which a high-latency DTN cannot sustain). The governing requirement is a clean split: *mesh messages are open to all, but internet egress is node-authorized*.

The naive model—an allow-list on the gateway—fails in a partition, because the gateway cannot reach whatever server provisions the list. Every comparable real-world system keeps the authority in a reachable server: cellular PCRF, captive-portal RADIUS, and enterprise zero-trust proxies [35]. Our contribution, ATTENUABLE EGRESS CAPABILITIES, instead puts the authority *in a token the requester presents*, following macaroons [4] and Biscuit [11] and building on the Ed25519 we already ship. A capability is a signed *grant* plus a signature-chain of narrowing *attenuations*, with three properties:

1. **Offline verification** — a gateway that trusts the issuer’s public key verifies the token with a signature check and a local clock; no server, works partitioned.
2. **Narrow-only delegation** — a holder appends a scope caveat (fewer hosts, tighter expiry, smaller byte cap) and passes the token on without contacting the issuer; the verifier conjoins every level, so a caveat can only *remove* authority.
3. **Unstrippable chain** — each attenuation is signed by the key the previous level delegated to and binds the previous signature, so caveats cannot be reordered, spliced, or stripped.

A SERVICE-CLASS label (`LifeSafety` $\subset$ `Messaging` $\subset$ `Interactive` $\subset$ `Bulk`) tiers the egress, the exit-side analogue of a 5G network slice or a DiffServ class [6, 18]; the `Messaging` tier is the zero-rated host-allowlist model familiar from in-flight Wi-Fi. The gateway additionally enforces SSRF defenses [31]: only `http/https`, and never loopback, private (RFC 1918), link-local, carrier-grade-NAT [44], or cloud-metadata targets, with the real fetcher required to re-check the resolved IP against DNS rebinding. Real-time calling (VoIP) is blocked *structurally*: a request/response DTN has no representation for a persistent bidirectional media stream.

In Lifeline. Crate `inet`. `ACCESSPOLICY` is a policy-decision point; `AllowList` is the trivial local issuer, `CapabilityPolicy` holds only a set of trusted issuer keys. Twenty tests cover forgery, splicing, stripping, expiry, subject-binding, monotone attenuation, offline verification across a simulated partition, and the SSRF and byte-ceiling guards.

11 The Residual Differential Transfer Framework

Several of Lifeline’s optimizations are instances of one idea: never send what the receiver can reconstruct from what it already has. Formally, to transfer a quantity Q when both parties share a reference R , transmit only the residual

$$\Delta = Q - \mathbb{E}[Q \mid R], \tag{1}$$

and let the receiver reconstruct $\hat{Q} = \mathbb{E}[Q \mid R] + \Delta$. Two dual regimes follow. In the *coding* regime R is shared exactly and Δ is a lossless delta (set reconciliation transmits $O(d \log |U|)$ for a symmetric difference of size d [27]; a manifest transmits only missing CIDs). In the *estimation* regime R is a shared predictor and Δ is a correction (differential time-sync transmits an offset against a stratum reference, not an absolute clock; differential GPS transmits a local correction against a common ephemeris). Recognizing these as one framework let us reuse a single mental model—and a single validity check against the information-theoretic literature—across time-sync, reconciliation, and content delta, rather than re-deriving each. We flag that the framework is a design lens, not a novel theorem; the underlying results are classical.

Threat	Mechanism in Lifeline	Residual limitation
Sybil [14]	no shared reputation to inflate; custody attribution is source-local (§5.3)	DHT population still Sybil-able; no proof-of-personhood
Eclipse [40, 19]	Kademlia incumbent retention + a per-endpoint diversity cap per bucket	id-space Sybil still possible; no proof-of-work node ids
Wormhole [20]	content integrity per CID; onion hides graph	no packet leashes; geographic spoofing possible
Black/grey-hole custody	source-attributed reputation: consecutive-miss demotion <i>and</i> a lifetime delivery-ratio rule for selective droppers (§5.3)	heuristic; adaptive adversary can sit at the ratio threshold
Spoofing / tampering	Ed25519 signatures inside the sealed envelope; per-CID integrity	key seizure defeats it
DoS / spam	priority-scaled Hashcash postage [2]; bounded buffers	sustained regional flooding
Info disclosure (content)	sealed-sender E2E; onion routing	gateway sees egress plaintext (by design)
Metadata / traffic analysis	sealed sender, onion, unlinkable bridge subkeys	<i>not</i> resistant to a global adversary [43, 33]
Egress abuse (SSRF)	scheme/host guards, resolved-IP recheck [31]	depends on the real fetcher honoring the contract
Token forgery / escalation	signed, chained, offline-verifiable capabilities (§10)	format needs independent review

Table 2: Threats, the mechanism addressing each, and the honestly-stated residual limitation. Implementation-level defenses (bounded CBOR decoding, decompression- bomb limits, zeroization, domain separation) are omitted for space.

12 Security Analysis

We analyze Lifeline against a taxonomy drawn from STRIDE [39] and the privacy dimensions of LINDDUN [12], plus the attacks specific to overlays and DTNs. Table 2 summarizes the threat, the mechanism, and its residual limitation.

The cautionary reference for this class of system is the break of Bridgefy [1], a deployed BLE mesh messenger that offered neither authenticity nor confidentiality and permitted tracking and impersonation; its failures directly motivate Lifeline’s sealed-sender design and its refusal to trust relays. We treat an independent cryptographic and protocol audit as a launch gate, not an afterthought.

13 Evaluation Methodology and Benchmarks

What we measure today. Lifeline ships an in-repo acceptance simulation held to a $\geq 95\%$ delivery target across partition, churn, and loss scenarios, plus ~ 200 unit and integration tests exercising each subsystem (crypto round-trips, custody, swarm fetch, capability verification). This establishes functional correctness and a delivery floor, but it is *not* a comparative performance evaluation.

Strategy	Delivery	Median lat.	Overhead	Mean hops	Peak buf.
Direct delivery (no relay)	0.0%	—	0.00	—	24
Binary spray-and-wait (default)	100.0%	90 s	10.0	2.75	240
Epidemic flooding	100.0%	90 s	24.0	2.75	576

Table 3: Mule-partition scenario (three radio-isolated clusters, one moving mule; seed 42, 700 ticks). Overhead = forwarded copies per delivered message; peak buf. = peak total bundles stored across all nodes. Direct delivery cannot cross the partition; spray-and-wait recovers epidemic’s full delivery at $2.4\times$ lower overhead *and* buffer. All delivered messages were offline-verified.

Strategy	Delivery	Median lat.	Overhead	Mean hops	Peak buf.
Direct delivery (no relay)	100.0%	220 s	2.0	1.00	48
Binary spray-and-wait (default)	100.0%	40 s	11.5	2.21	276
Epidemic flooding	100.0%	30 s	46.0	4.04	1104

Table 4: Random Waypoint scenario (24 nodes, seed 42, 700 ticks). A connected setting: direct delivery reaches everyone eventually but slowly; epidemic is marginally fastest at $4\times$ spray-and-wait’s buffer and overhead.

An in-repo comparative baseline. We ship a reproducible harness (`sim::bench`) that runs the *same* partitioned scenario—three radio-isolated clusters bridged by a single moving data mule, with every phone messaging counterparts one and two clusters away—under three forwarding strategies, using the DTN standard metric set: delivery ratio, delivery latency, and overhead (relay) ratio [41, 42]. Table 3 reports the result (seed 42, 700 ticks). It reproduces the expected ordering exactly: no-relay *direct delivery* cannot cross a partition and delivers nothing; *epidemic* flooding [42] delivers everything but at the highest overhead; and Lifeline’s default *binary spray-and-wait* [41] matches epidemic’s 100% delivery and latency while using $2.4\times$ fewer transmissions per delivered message (10.0 vs. 24.0)—the whole point of controlled replication. Every message also carries an offline-verifiable delivery proof (100% verified).

A Random-Waypoint run. The harness also drives contacts from a mobility model instead of clusters. Table 4 is the canonical Random Waypoint setting [7] (24 nodes roaming a square, in contact within radio range). Here the network is intermittently *connected*, so even no-relay direct delivery eventually reaches 100%—but at $5.5\times$ spray-and-wait’s latency. Epidemic shaves a little more latency (30 vs. 40 s) but pays $4\times$ the buffer and overhead. Spray-and-wait is again the knee of the curve: near-epidemic latency at a quarter of the cost.

What remains. The harness now reports the full metric set (Table 5: delivery, latency, overhead, hop count, buffer occupancy) under both cluster and mobility-driven contacts, and it replays an explicit contact trace via a `TraceMobility` driver. The remaining empirical work is to feed it *real* traces at scale—the CRAWDAD `cambridge/haggle` Bluetooth traces [37] and MIT Reality Mining [15] (not vendored here for licensing reasons)— and to cross-check against the reference ONE simulator [21] and the Working Day Movement Model [16].

Metric	Definition
Delivery ratio	fraction of unique messages delivered within TTL
Delivery latency	creation-to-delivery time (median and CDF)
Overhead ratio	(relayed – delivered)/delivered
Buffer occupancy	per-node storage over time
Transmissions/delivery	energy proxy for battery-limited nodes
Swarm completion time	rounds to complete a k -block object vs. # providers

Table 5: The evaluation metric set, following the DTN benchmarking literature [41, 21]. All but swarm-completion time are now reported by `sim::bench`.

14 Related Work

Lifeline sits at the confluence of DTN routing [9, 42, 41], content-addressed storage [3, 10], secure messaging [22, 32], peer-to-peer overlays [23], and capability-based authorization [4, 11]. It differs from deployed mesh messengers in refusing to trust relays (cf. the Bridgefy break [1]), from Mesh-tastic [25] in providing end-to-end sealed-sender confidentiality and a routing/custody layer rather than a flood, and from server-anchored systems in keeping identity, routing, and *authorization* all functional under partition. Its anonymity goals are weaker than mixnets [43, 33], a trade we make deliberately for latency and offline operation.

15 Limitations and Future Work

- **Independent review.** No external cryptographic/protocol audit has been done; the capability token format and the forward-secrecy claim need it.
- **Comparative evaluation.** A study on standard traces (§13) against epidemic/spray-and-wait baselines is outstanding.
- **Egress quota and revocation.** Capabilities enforce a per-response byte ceiling but not yet a cumulative token-bucket quota, and rely on short expiry rather than revocation lists.
- **Metadata protection.** Traffic-analysis resistance is limited; integrating cover-traffic or mixing [33] is future work.
- **Overlay hardening.** The DHT now caps contacts per endpoint per bucket (an eclipse mitigation) and custody attribution now catches grey-hole (selective) droppers via a delivery-ratio rule; proof-of-work node ids and adaptive-adversary resistance remain future work.
- **Live wiring.** Time-sync, reconciliation, and geocast exist and are tested as crates but are not yet all driven by the live node.

16 Conclusion

Project Lifeline shows that a messenger can keep its identity, its delivery guarantee, its confidentiality, and even its *authorization decisions* working when every server is gone—by moving each of these into the endpoints and into the data itself. Content addressing gives integrity and dedup for free; a manifest turns “complete” into a checkable predicate; swarm fetch turns every peer and every carrier into a source; and attenuable capabilities turn internet egress into a portable, offline-

verifiable right. The remaining work is empirical and adversarial: measure it against the field’s benchmarks, and subject its cryptography to independent attack.

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